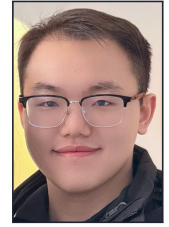


Zhiyuan Ouyang

2nd-year Ph.D. @ East China Normal University / Shanghai Institute of Intelligence

✉ zhiyuanouyang@sii.edu.cn 🌐 ChiryanOY

🐦 Chiryan_OY 🌐 chiryanoy.github.io



Summary

My research centers on **Generative Models**, focusing on model generalization and underlying topologies to enhance generation efficiency and controllability. In AI engineering, I am experienced in large-scale cluster scheduling via **Slurm** and **DDP**, with hands-on expertise in concurrent training and tuning on **256 * H200 GPUs**. I am also proficient in building and deploying customized visual generation pipelines based on **ComfyUI**. Furthermore, I have practical experience in developing **Web3** infrastructure and **on-chain execution systems** for complex strategy deployment.

Education

ECNU/SII 2nd-year Ph.D. in Computational Mathematics

Shanghai, China

Advisor: Prof. Junchi Yan

Honors & Awards: National Scholarship for PhD Students, China (Top 1.5%)

Competition Awards: C/C++ Programming Contest, Collegiate Division (Provincial First Prize, Top 5%)

Research Interests: Diffusion Generative Models / Time Series / AIGC / AI for Trading

Publications

Zhiyuan Ouyang (First Author). Primal-Spectral Generative Modeling: Fast Analytical Generation via Pseudoinverse Lévy Inversion. *Published in ICML 2026 (CCF-A)*

Zhiyuan Ouyang (First Author). A Novel Network for Resolving Subjective Masking Differences and Accurate Thyroid Nodule Diagnosis. *Published in Computational Biology and Chemistry (JCR-Q2)*

Zhiyuan Ouyang (First Author). Model Detection for Grey Forecasting Model with Polynomial Term. *Published in Communications in Statistics (JCR-Q3)*

Projects & Experience

Generative Modeling in Characteristic Function Space (Outstanding Ph.D. Project) 📄 **PaperNov** 2025 – May 2026

- **Theoretical Innovation:** Explored the evolution mechanism from Primal Data Distribution to Characteristic Function. Transformed the distribution into a continuous function via Spectral Methods, solving the challenge of neural networks reconstructing global topology from sparse samples, and providing theoretical guarantees for convergence.
- **Algorithm Architecture:** Proposed the **PriSpecNet** architecture and designed a single-step evaluation (1-NFE) Pseudoinverse Lévy Inversion (PiLI) solver. This transforms the generation process into a Closed-form Analytic problem, completely eliminating the reliance on iterative Integration while maintaining full compatibility with Stochastic Interpolation.
- **Application Performance:** Unified generation and forecasting in the time series domain, reducing Context-FID metrics on Sines, Solar, ETTh, and Stock datasets by 50.0%, 41.5%, 80.6%, and 63.1%, respectively. In the ImageNet 256×256 generation task, achieved an FID of 1.66 with only **26 Gflops** computational overhead, realizing a **170x** computation reduction compared to 25-NFE DPM-Solver++.

Baizhi Xunlian Technology Co., Ltd. (Head of AI Department)

Sep 2023 – Sep 2024

- **Generative Model R&D:** Led the construction of visual generation Pipelines based on **ComfyUI** and **Stable Diffusion**. Conducted Data Processing and model Fine-tuning for specific vertical domains, delivering highly controllable, End-to-End customized image generation solutions.
- **Engineering Deployment:** Responsible for architecture planning and On-Premises deployment of enterprise-grade AI production environments. Performed Inference Optimization for heterogeneous computing platforms to break through edge deployment bottlenecks, ensuring High Availability of core businesses.
- **Team Management:** Comprehensively managed the technical evolution and Agile R&D of the AI department. Provided AIGC commercial landing consulting for enterprise clients; established internal engineering standards, led technical training, and drove efficient project delivery by Cross-functional teams.

MimicPolymarket: On-chain Router (Independent Open Source) 🌐 **GitHub**

Feb 2026 – Present

- **Core Architecture:** Designed an industrial-grade execution engine with concurrent multi-wallet routing across EOA / Safe / Deposit Wallet runtimes. Deeply integrated Polymarket's Relayer routing and the POLY_1271 signature flow to enable Account Abstraction (AA), **living account** achieved **352% monthly PnL**.
- **Risk Control & Execution:** Built a per-trader quantitative risk matrix covering mimic ratio, per-order / position caps, buy / sell slippage, and aggregation windows. Buy-side uses balance-aware Limit Orders; sell-side performs Best Bid anchored FOK execution with dynamic scaling out, reducing Market Impact and abnormal fills.
- **Signal Aggregation & Resilience:** Designed a fixed-window Order Aggregation engine that batches fragmented Snipe/MEV signals by trader / market / asset / side and converts them into synthetic trades via weighted-average pricing; the full lifecycle is persisted to MongoDB, with serial task queues, retry recovery, and state replay enabling a low-latency, highly available, auditable on-chain trading loop.

Technical Skills

Programming Languages: [Python](#) - focused on analytical layer R&D like algorithm modeling and data exploration; [TypeScript](#) / [Rust](#) / [C++](#) - leading high-performance execution layer implementation, proficient in utilizing the Vibe Coding paradigm to rapidly build multi-language architectures.

Deep Learning & Generative Models: Proficient in [PyTorch](#) and [JAX](#) for core algorithm R&D; built large-scale visual generation architectures based on Diffusers; highly experienced in Slurm cluster scheduling and DDP distributed training (with concurrent training and performance tuning experience on up to 256* H200 GPUs), frequently using Tensorboard for experiment tracking.

Web3 & On-chain Execution: Built on-chain execution systems with [Ethers.js](#) / [Viem](#) and the Polymarket CLOB SDK; proficient in Account Abstraction (AA), Deposit Wallet routing, Relayer integration, FOK / Limit execution, slippage protection, and trade aggregation mechanisms; capable of complex contract interactions, state monitoring, and low-latency copy-trading architecture design.

Architecture & Engineering: Applied dynamic state machine patterns to design robust risk control execution flows; used [MongoDB](#) for low-latency persistence of high-frequency trading data; proficient in [Docker](#) containerized deployment and Jest-based test-driven development.